Bridge Circuits

DC WHEATSTONE BRIDGES

Figure 2-1 (a) shows the classic de Wheatstone bridge circuit. The circuit is redrawn in Fig. 2-1 (b) to more clearly show the fact that we may consider this circuit as two resistor voltage dividers connected in parallel across a voltage source E. Current I5 through the galvanometer is given by

$$I5 = \frac{E_0}{R5 + R_m} \tag{2-1}$$

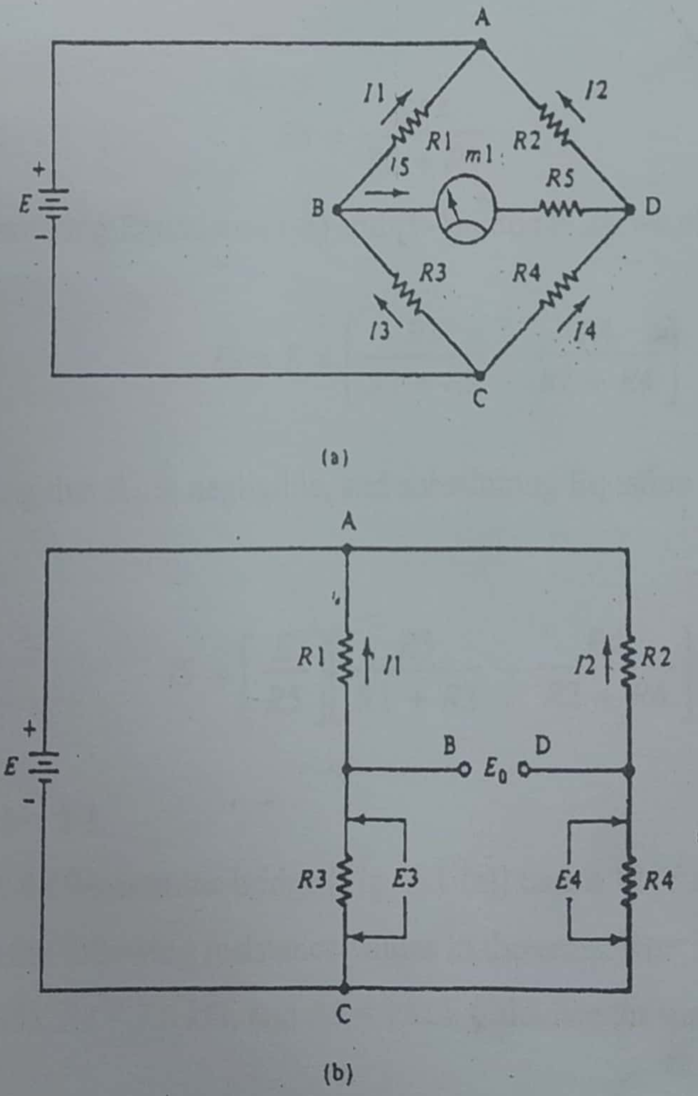


Fig. (2-1) (a) The Wheatstone bridge, (b) redrawn to make more clear.

where I5 = the current flowing in meter *Ml* in *amperes* (A)

E_0 = the potential in. *volts* (V) between points B and D

R_5 = the resistance of resistor *R5* in ohms (Ω)

R_m = the resistance of the meter movement in ohms (Ω).

Voltage E_0 is the difference between voltages E_3 and E_4:

$$E_0 = E3 - E4 \quad (2\text{-}2)$$

But E3 and E4 are given in terms of E and the resistances:

$$E3 = \frac{ER3}{R1 + R3} \quad (2\text{-}3)$$

And

$$E4 = \frac{ER4}{R2 + R4} \quad (2\text{-}4)$$

By substituting Equations (1-3) and (1-4) into (1- 2), we obtain

$$E_0 = E \times \left[\frac{R3}{R1 + R3} - \frac{R4}{R2 + R4}\right] \quad (2\text{-}5)$$

Assuming that R_m is negligible, and substituting Equation (2-5) into (2-1), we have

$$I5 = \left[\frac{E}{R5}\right]\left[\frac{R3}{R1 + R3} - \frac{R4}{R2 + R4}\right] \quad (2\text{-}7)$$

Example : 2-1

A de Wheatstone bridge [Fig. 2-1 (a)] uses a 12-V battery for excitation and has the following resistance values in the arms: R_1 = 1.2 kΩ, R_2 = 1.5 kΩ, *R3* = 4 kΩ, *R4* = 3.6 kΩ, and *RS* = 1 kΩ. Calculate the meter current.

Solution :

$$I5 = \frac{E}{R5}\left[\frac{R3}{R1 + R3} - \frac{R4}{R2 + R4}\right]$$

$$= \frac{12}{1000}\left[\frac{4}{(1.2 + 4)} - \frac{3.6}{(1.5 + 3.6)}\right]$$

$$= \frac{(12)(0.77 - 0.71)}{1000} = 7.2 \times 10^{-4}\ \text{A} = \mathbf{0.72\ mA}$$

BRIDGES IN THE NULL CONDITION

The **null** condition in a Wheatstone bridge exists when the output voltage E_0 is zero. We may find the sole necessary condition for null by setting Equation (2-6) equal to zero. Thus,

$$\left[\frac{E}{R5}\right]\left[\frac{R3}{R1 + R3} - \frac{R4}{R2 + R4}\right] = 0 \qquad (2\text{-}7)$$

For Equation (2-7) to evaluate to zero. either the quantity E/R5 or the quantity inside the braces must be zero. The former quantity is never zero in practical situations, so we may conclude that

$$\frac{R3}{R1 + R3} - \frac{R4}{R2 + R4} = 0 \qquad (2\text{-}8)$$

From which we may also conclude that:

$$\frac{R3}{R1 + R3} = \frac{R4}{R2 + R4} \qquad (2\text{-}9)$$

Equation (2-9) can be simplified by the following steps. At null, E_0 is zero, so I5 is also zero. Therefore;

$$I1 = I3 \qquad (2\text{-}10)$$

$$I2 = I4 \qquad (2\text{-}11)$$

We may then write

$$I1R1 = I2R2 \qquad (2\text{-}12)$$

and

$$I1R3 = I2R4 \qquad (2\text{-}13)$$

The sole necessary condition for bridge null is obtained by dividing Equation (1-13) into Equation (1-12), which yields Equation (1-14):

$$\frac{R1}{R3} = \frac{R2}{R4} \qquad (2\text{-}14)$$

From Equation (2- 14) we can compute any resistance value if the other three values are known. Note that it is not necessary for the resistor values to be equal-only that the ratios R_1/R_3 and R_2/R_4 be equal. Equation (2-14) describes the Wheatstone bridge null condition and is essential to understanding many different classes of electronic instrumentation circuits.

Example (2-2)

A Wheatstone bridge has the following arm values: R_1 = 1.6 k.Ω, R_2 = 500 Ω, R3 is unknown, and R4 = 1800 Ω. What value of R_3 brings the bridge into the null condition?

Solution

$$R3 = R1R4/R2 \qquad \text{[(from Eq. (2-14)]}$$

$$= \frac{(1600)(1800)}{500} = 5760\ \Omega$$

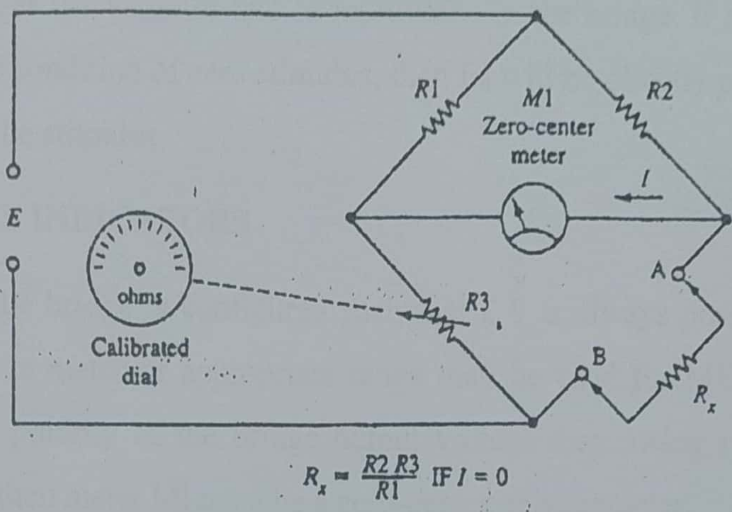


Fig. (2-22) Measuring resistance with a Wheatstone bridge.

DC BRIDGE APPLICATIONS

Equation (2-14) gives us a means for measuring resistances very accurately-far more accurately, in fact, than is possible on a volt-ohm-milliammeter or even most digital ohmmeters. In Fig. 2-2, the unknown resistance Rx is given by

$$R_x = \frac{R2R3}{R1} \qquad (2\text{-}15)$$

(when the bridge is in null).

If we make R_1 and R_2 fixed and link potentiometer *R3* to a dial that is calibrated in ohms, then we may measure Rx· We could, for example, make R_1 equal to *R2,* so the ratio R_2/R_1 is unity. In that case Rx will equal *R3* at the null point. To measure Rx, connect it to the bridge terminals A and B, and adjust R_3 for a *zero* reading on meter *Ml.* In some commercial bridges, resistors R_2/R_1 serve as a range multiplier and have different values on different resistance ranges.

Another category of wheatstone bridge applications begins with a balanced bridge and then measures the *amount* of unbalance when one or more of the resistance arms *changes.* This principle is used in many *transducers* in which a stimulus parameter, such as force, temperature, displacement, and so on, changes the value of one or more arms in the bridge. If all arms are equal under the condition of zero stimulus, then E_0 will be directly proportional to the value of the stimulus.

DC NULL INDICATORS

If the bridge is configured such that £ 0 is always positive or negative, then any de meter of appropriate range may be used for MI. If, on the other hand, the polarity of the bridge output voltage may swing either positive or negative, then meter Ml must be a zero-center galvanometer.

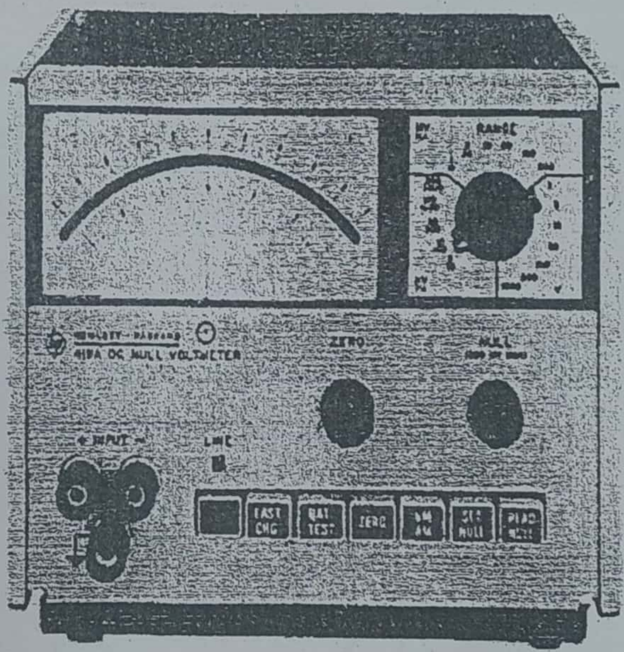


Fig. (2-3) Zero-center null voltmeter (courtesy of Hewlett-Packard).

Interestingly enough, either a voltmeter or a current meter (milli- or microammeter) may be used for M_1. In bridges that are passive, a current meter is generally used. Typical ranges fall within the 0-100-µA to 0-1-mA range. The smaller full-scale current movements are capable of better resolution, provided that the minimum change of resistance step used to null the bridge produces current changes that are considerably less than the full-scale current of the meter movement. Figure 2-3 shows a battery-operated de null meter that is capable of detecting E_0 values down to the microvolt range. This instrument is a very sensitive amplified voltmeter that is a special case of the instruments discussed later in this book.

AC BRIDGES

The dc Wheatstone bridge will work with either ac or de excitation, although in the for- mer case the null detector must be a device that is sensitive to ac. Certain other types of bridge circuits are designed specifically for ac use and have inductive or capacitive reactances in one or more of the bridge arms.

AC NULL DETECTORS

The dc galvanometer used with the resistive Wheatstone bridge will not respond to ac, so it cannot be used directly as the null detector in ac bridges. Many rectifier ac meters based on the de movement are not sensitive enough to indicate the null precisely, so these meters are limited to applications where a relatively shallow (i.e., broad) null can be tolerated.

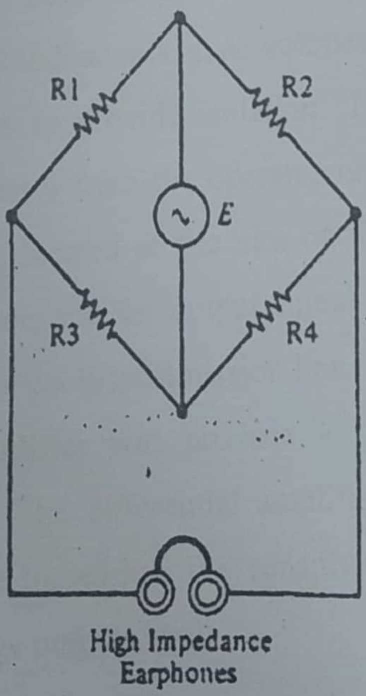


Fig. (2-4) ac excited Wheatstone bridge using headphones as the null detector.

Alternating current electronic voltmeters and oscilloscopes, on the other hand, are quite sensitive and thus can indicate the ac null with great precision. These instruments can be used to replace the de galvanometer-rectifier combination where sensitive measurements are to be made.

In some cases, high-impedance earphones are used as a null detector (see Fig. 2-4) in ac bridges. A surprising degree of sensitivity is possible if the operator has high hearing acuity. In general, highly efficient communications *(not* high-fidelity) earphones with an impedance of 1000 Ω or higher are used for this purpose.

Problems sometimes occur when one is trying to use instruments with ground-referenced inputs as a null detector, and these problems are especially troublesome if the ac excitation source is also grounded.

There are two basic approaches to solving this problem. One is transformer coupling to the null detector (thereby isolating the detector from the bridge), and the other is the use of a differential amplifier. Figure 2-5(a) shows the use of a transformer to couple the signal from the E_0 nodes of the bridge to an external null indicator such as an oscilloscope or ac voltmeter. In this particular example the null indicator is an ac voltmeter. Transformer coupling is often used in RF bridges to provide isolation. The bridge components are sensitive to body capacitance from the operator or other nearby disturbances. The bridge then may be located at the site of the measurement in relative isolation from the operator, and the output signal may be brought to the null indicator through a coaxial cable transmission line.

A differential amplifier will produce a single-ended output from a differential input signal. The differential amplifier is useful for isolation, as described above, or for increasing the sensitivity of the null indicator by amplification of the bridge output signal.

Many oscilloscopes can be equipped with a differential amplifier plug-in, while even most nonplug-in models can be used in a quasi-differential mode if they are equipped with two channels. To use a two-channel oscilloscope in quasi-differential, use the two inputs. as inputs to a differential amplifier, and set the oscilloscope's mode switch in A-B.

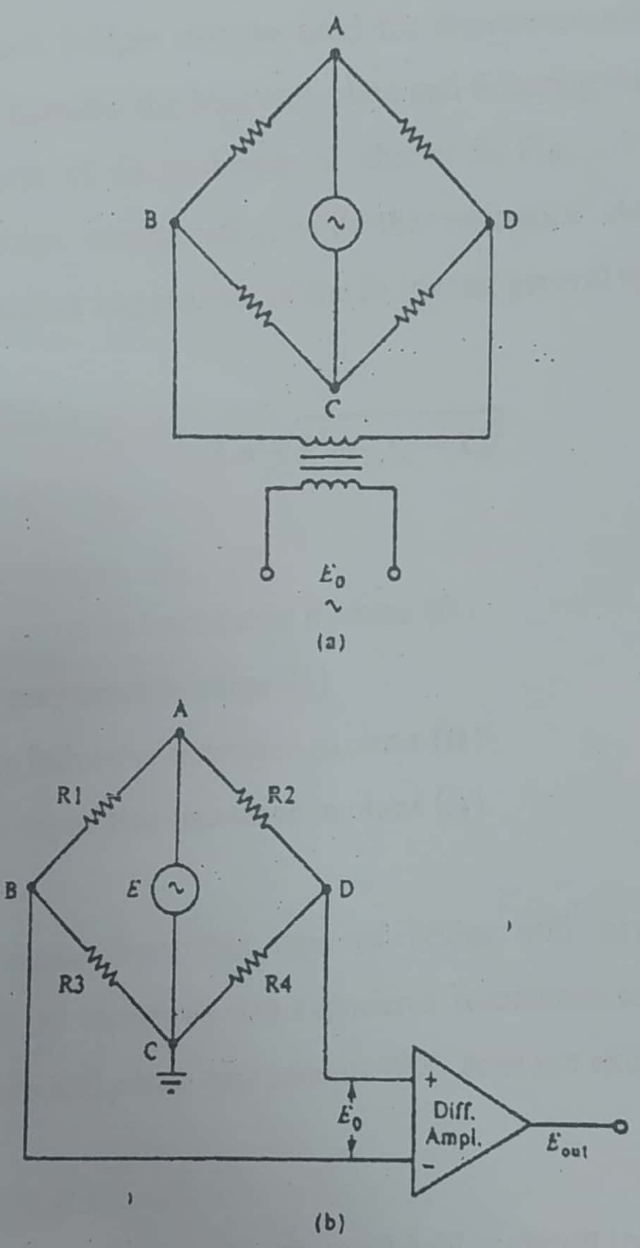


Fig. (2-5) (a) Transformer-coupled ac Wheatstone bridge, (b) differential amplifier output for an ac Wheatstone bridge.

TYPES OF AC BRIDGES

Although the resistive Wheatstone bridge is technically an "ac bridge" if the excitation source is ac, we usually limit the designation to those bridge types that use a reactive element such as a capacitor or an inductor in one or more bridge arms. Such bridges may be used for *impedance* measurements. In this chapter we will consider the Maxwell, Hay, and Schering bridge circuits.

The general form of an ac bridge is shown in Fig. 2-7; it is the standard Wheatstone bridge configuration with the resistance elements in the arms replaced by complex impedances of the following general type:

$$Z = \sqrt{R^2 + (X_L - X_c)^2} \qquad (2\text{-}16)$$

where Z = the complex impedance in ohms (Ω)

R = the resistance in ohms (Ω)

X_L = the inductive reactance in ohms (Ω)

X_c = the capacitive reactance in ohms (Ω)

The output signal from this type of bridge will have only a' *magnitude* component if the inductive and capacitive reactances cancel, but it will have both *magnitude* and *phase* components if X_L does not exactly balance X_c.

MAXWELL'S BRIDGE

Figure 2-8(a) shows the Maxwell bridge circuit in which Z_2 and Z_3 of the general configuration have been replaced by pure resistances, Z_1 has been replaced by a parallel combination R_1/C_1, and Z_4 by the series combination L_1/R_4.

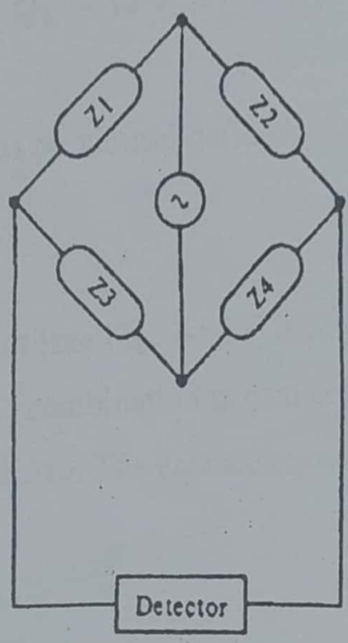


Fig. (2-7) Generalized circuit for complex ac bridges.

The null condition for a Maxwell bridge circuit is given by the following equations:

$$L1 = R2 \times R3 \times C1 \quad (2\text{-}17)$$

$$R4 = \frac{R2 \times R3}{R1} \quad (2\text{-}18)$$

The Maxwell bridge is often used to measure unknown values of inductance (e.g., L_1) because the balance equations are totally *independent* of frequency. The bridge is also not too sensitive to resistive losses in the inductor. Additionally, it is much easier; to obtain easily calibrated capacitor standards for C_1 than it is to obtain inductor standards for L_l. As a result, the principal use of the Maxwell bridge is the measurement of inductances. The Maxwell circuit is often used in *Q-meters,* that is, instruments that measure the quality factor Q of inductors. The equation for Q, however, is frequency sensitive:

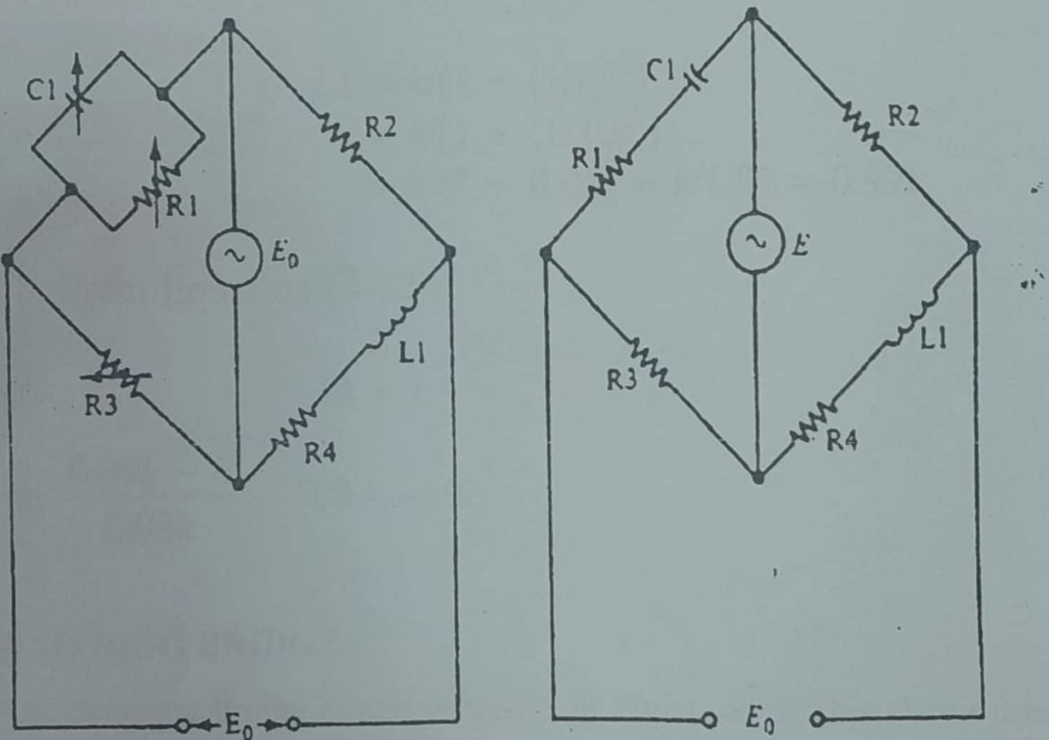
$$Q_L = \omega \times R1 \times C1 \tag{2-19}$$

where $\omega = 2\pi f$

R_1 and C_1 are as previously defined.

THE HAY BRIDGE

The Hay bridge circuit [see Fig. 2-8(b)] is physically similar to Maxwell's bridge, except that the R_1/C_1 combination is connected in series. The Hay bridge is, however, frequency sensitive. The balance equations are as follows:



$$L1 = R2\,R3\,C1$$

$$R4 = \frac{R3}{R1} \times R2$$

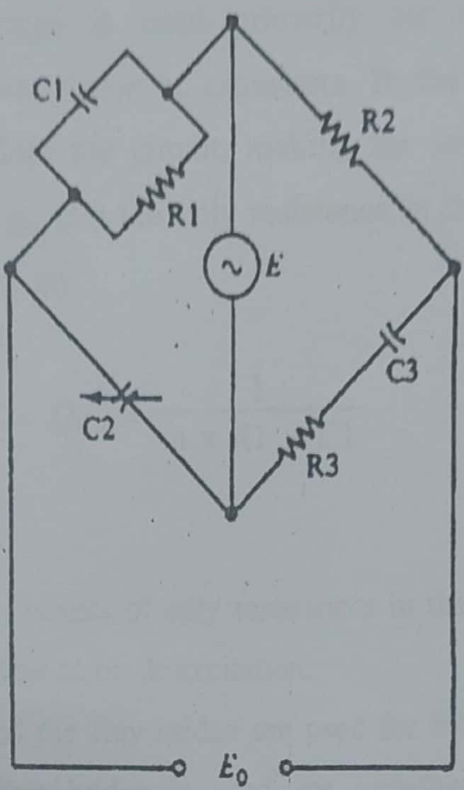
$$Q_L = \omega\,C1\,R1$$

(a)

$$R4 = \frac{R2\,R3}{R1} \times \frac{1}{Q^2+1} \qquad L1 = \frac{R2\,R3\,C1}{1+\left(\frac{1}{Q}\right)^2}$$

$$Q = \frac{1}{\omega\,R1\,C1}$$

(b)



$R3 = \frac{C1}{C2} \times R2 \qquad C3 = \frac{R1}{R2} \times C2$

(c)

Fig. (2-8) (a) Maxwell's bridge, (b) the Hay bridge, (c) the Schering bridge.

$$L1 = \frac{R2 \times R3 \times C1}{1 + \left[\frac{1}{Q}\right]^2} \tag{2-20}$$

$$R4 = \left[\frac{R2 \times R3}{R1}\right] \times \left[\frac{1}{Q^2 + 1}\right] \tag{2-21}$$

$$Q = \frac{1}{\omega \times R1 \times C1} \tag{2-22}$$

The Hay bridge is preferred for measuring inductances with high Q figures, while Maxwell's bridge works best with low-Q inductors.

Note that a frequency-independent version of Equation (2-20) is possible when the Q is very high-that is, greater than about 10-in which case Equation (2-20) can be rewritten in the approximation form:

$$L_1 = R_2 \text{ X } R_3 \text{ X } C_1 \tag{2-23}$$

The error caused by using the approximation of Equation (2-23) is 1% if the Q is .10 and 0.01% if the Q is 100.

Proof" Let .

$$R_2C_3C_1 = k = \text{constant}$$

Assume

$$Q = 10$$

a. From Equation (2-20):

$$\begin{aligned} L1 &= k/[1 + (1/Q)^2] \\ &= k/[1 + (1/10)^2] \\ &= k/(1 + 0.01) = k/1.01 = 0.99k \end{aligned}$$

b. From Equation (2-22):

$$L1 = k$$

c. $\dfrac{0.99k - k}{0.99k} \times 100 = -1\%$

THE SCHERING BRIDGE

The Schering bridge circuit is shown in Fig. 1 -8(c). This circuit uses a parallel RC network R_1/C_1 for Z_1, a resistance R_2 for Z_2, a capacitive reactance C_2 for Z_3, and a series R_C network R_3/C_3 for Z_4. The Schering bridge balance equations are

$$R3 = \frac{R2 \times C1}{C2} \tag{2-24}$$

$$C3 = \frac{C2 \times R1}{R2} \tag{2-25}$$

The Schering ·bridge is used primarily for the measurement of *capacitance* and the *power factor* of capacitors. In the latter application no actual *R3* is connected into the circuit, making the series resistance of the capacitor being tested (e.g., C3) the only resistance in that arm of the bridge. The capacitor *Q* is found from

$$Q_{C3} = \frac{1}{\omega \times R1 \times C1} \qquad (2\text{-}26)$$

Summary:

1. The Wheatstone bridge consists of only resistances in the respective branches; it may be used with either ac or .de excitation.
2. The Maxwell bridge and the Hay bridge are used for inductance measurements, whereas the Schering bridge is used for capacitance measurement. The Wheatstone bridge is used mainly for resistance measurements.
3. The zero-center galvanometer is· used as a null detector in de Wheatstone bridges;
4. Alternating-current current meters or ac voltmeters are used as null detectors in ac bridges when there are no phase components in the output signal, while an oscilloscope or a phase detector can be used when there are both phase and magnitude components.

Recapitulation

Now go back and try to answer the questions at the beginning of the chapter. when you are finished, answer the questions and work the problems below. Place a mark beside each problem or question that you cannot answer, and then go back and reread appropriate sections of the text.

QUESTIONS

1. Draw the circuit for a dc Wheatstone bridge and label the components (R_1 , R_2, R_3, etc.).
2. Using the component designations given in Question1, write the *null condition* equations.
3. Draw the circuit for a *Maxwell* bridge, label the components, and write the null condition equations.
4. Draw the circuit for a *Hay* bridge, label the components, and write the null condition equations.
5. Draw the circuit for a Schering bridge, label the components, and write the null condition equations.
6. What type of ac null detector is best suited for (a) the Wheatstone bridge, (b) the Hay bridge, (c) the Schering bridge, and (d) the Maxwell bridge?
7. What type of bridge is best suited for measuring capacitance?
8. What type of bridge is best suited for measuring the inductance of high- Q coils?
9. What type of bridge ,is best suited for measuring the inductance of low-Q coils?
10. What type of bridge is best suited for measuring the power factor of a capacitor?
11. A zero-center galvanometer with a range of 1-0-1 mA is used as the null detector in a dc Wheatstone bridge. What type of circuit or device may be used to increase the sensitivity of the galvanometer so that 10 μA will deflect the meter full scale?
12. What type of device or circuit may be used with an ac Wheatstone bridge to drive a ground-referenced null indicator?

Problems :

1. Is the Wheatstone bridge of Fig. 2-9 balanced? What is the value of *E0?*
2. *R4* in Fig. 2-9 is changed to 5.2 ill. What is the value of E_0?
3. A 20-kΩ resistor is connected between points A and B in Fig. 2-9, and *R4* is changed to 4.7 kΩ. Calculate the current through the 20-kΩ resistor.
4. What value would E_o in Fig. 2-9 become if *R2* were shunted by a 1 00-kn resistor?

5. An unknown resistance Rx is connected into Fig. 2-9 in place of R_4, and $R3$ is adjusted to reestablish the null condition. What is the value of Rx if R3 has a value of 576 Ω?

6. What is the value of $L1$ in Fig. 2-10?

7. What is the value of $R4$ in Fig. 2-10?

8. Find the Q of the $R4/L1$ network in Fig. 2-10

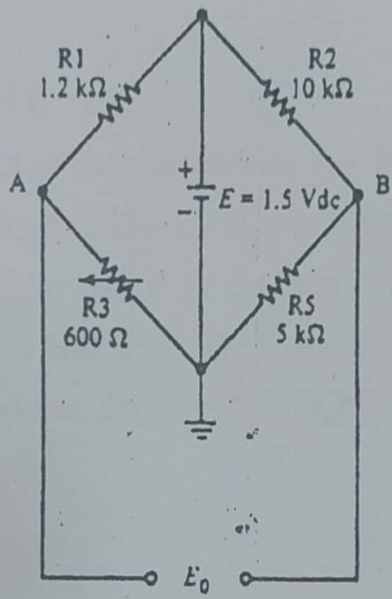


Fig. (2-9) Wheatstone bridge for Problems 1-5.

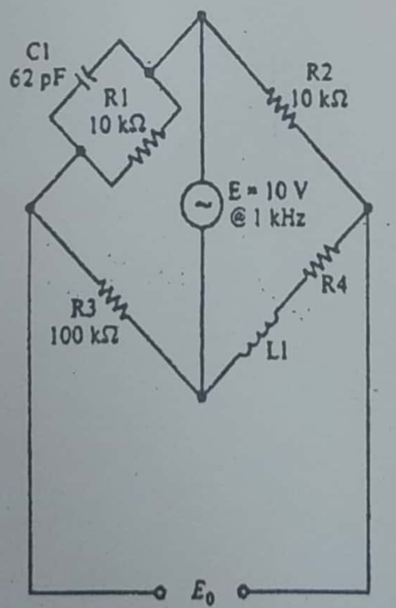


Fig. (2-10) Network for Problems 6-10.

9. A Hay bridge is formed by using the same components as in Fig. 2-10. An unknown inductor is connected in place of Ll, and the bridge is adjusted to null. At that point it is found that Cl = 93 pF. What is the value of the unknown inductor?

10. What is the *Q* of the coil in Problem 9?

11. Use the loop method of network analysis to derive the expression for E_0 in an unbalanced de Wheatstone bridge.

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